

Evaluate the product $\prod_{k=2}^n (1 - 1/k^2)$.

A.2 Bounding summations

You can choose from several techniques to bound the summations that describe the running times of algorithms. Here are some of the most frequently used methods.

Mathematical induction

The most basic way to evaluate a series is to use mathematical induction. As an example, let's prove that the arithmetic series $\sum_{k=1}^n k$ evaluates to $n(n+1)/2$. For $n = 1$, we have that $n(n+1)/2 = 1 \cdot 2/2 = 1$, which equals $\sum_{k=1}^1 k$. With the inductive assumption that it holds for n , we prove that it holds for $n+1$. We have

$$\begin{aligned}\sum_{k=1}^{n+1} k &= \sum_{k=1}^n k + (n+1) \\ &= \frac{n(n+1)}{2} + (n+1) \\ &= \frac{n^2 + n + 2n + 2}{2} \\ &= \frac{(n+1)(n+2)}{2}.\end{aligned}$$

You don't always need to guess the exact value of a summation in order to use mathematical induction. Instead, you can use induction to prove an upper or lower bound on a summation. As an example, let's prove the asymptotic upper bound $\sum_{k=0}^n 3^k = O(3^n)$. More specifically, we'll prove that $\sum_{k=0}^n 3^k \leq c3^n$ for some constant c . For the initial condition $n = 0$, we have $\sum_{k=0}^0 3^k = 1 \leq c$ as long as $c \geq 1$. Assuming that the bound holds for n , we prove that it holds for $n+1$. We have

$$\begin{aligned}\sum_{k=0}^{n+1} 3^k &= \sum_{k=0}^n 3^k + 3^{n+1} \\ &\leq c3^n + 3^{n+1} \quad (\text{by the inductive hypothesis}) \\ &= \left(\frac{1}{3} + \frac{1}{c}\right) c3^{n+1} \\ &\leq c3^{n+1}\end{aligned}$$

as long as $(1/3 + 1/c) \leq 1$ or, equivalently, $c \geq 3/2$. Thus, $\sum_{k=0}^n 3^k = O(3^n)$, as we wished to show.

You need to take care when using asymptotic notation to prove bounds by induction. Consider the following fallacious proof that $\sum_{k=1}^n k = O(n)$. Certainly, $\sum_{k=1}^1 k = O(1)$. Assuming that the bound holds for n , we now prove it for $n + 1$:

$$\begin{aligned}\sum_{k=1}^{n+1} k &= \sum_{k=1}^n k + (n+1) \\ &= O(n) + (n+1) \quad \leftarrow \text{wrong!} \\ &= O(n+1) .\end{aligned}$$

The bug in the argument is that the “constant” hidden by the “big-oh” grows with n and thus is not constant. We have not shown that the same constant works for *all* n .

Bounding the terms

You can sometimes obtain a good upper bound on a series by bounding each term of the series, and it often suffices to use the largest term to bound the others. For example, a quick upper bound on the arithmetic series (A.1) is

$$\begin{aligned}\sum_{k=1}^n k &\leq \sum_{k=1}^n n \\ &= n^2 .\end{aligned}$$

In general, for a series $\sum_{k=1}^n a_k$, if we let $a_{\max} = \max \{a_k : 1 \leq k \leq n\}$, then

$$\sum_{k=1}^n a_k \leq n \cdot a_{\max} .$$

The technique of bounding each term in a series by the largest term is a weak method when the series can in fact be bounded by a geometric series. Given the series $\sum_{k=0}^n a_k$, suppose that $a_{k+1}/a_k \leq r$ for all $k \geq 0$, where $0 < r < 1$ is a constant. You can bound the sum by an infinite decreasing geometric series, since $a_k \leq a_0 r^k$, and thus

$$\begin{aligned}\sum_{k=0}^n a_k &\leq \sum_{k=0}^{\infty} a_0 r^k \\ &= a_0 \sum_{k=0}^{\infty} r^k\end{aligned}\tag{A.15}$$

$$= a_0 \frac{1}{1-r} .\tag{A.16}$$

You can apply this method to bound the summation $\sum_{k=1}^{\infty} (k/3^k)$. In order to start the summation at $k = 0$, rewrite it as $\sum_{k=0}^{\infty} ((k+1)/3^{k+1})$. The first term (a_0) is $1/3$, and the ratio (r) of consecutive terms is

$$\begin{aligned}\frac{(k+2)/3^{k+2}}{(k+1)/3^{k+1}} &= \frac{1}{3} \cdot \frac{k+2}{k+1} \\ &\leq \frac{2}{3}\end{aligned}$$

for all $k \geq 0$. Thus, we have

$$\begin{aligned}\sum_{k=1}^{\infty} \frac{k}{3^k} &= \sum_{k=0}^{\infty} \frac{k+1}{3^{k+1}} \\ &\leq \frac{1}{3} \cdot \frac{1}{1-2/3} \\ &= 1 .\end{aligned}$$

A common bug in applying this method is to show that the ratio of consecutive terms is less than 1 and then to assume that the summation is bounded by a geometric series. An example is the infinite harmonic series, which diverges since

$$\begin{aligned}\sum_{k=1}^{\infty} \frac{1}{k} &= \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{k} \\ &= \lim_{n \rightarrow \infty} \Theta(\lg n) \\ &= \infty .\end{aligned}$$

The ratio of the $(k+1)$ st and k th terms in this series is $k/(k+1) < 1$, but the series is not bounded by a decreasing geometric series. To bound a series by a geometric series, you need to show that there is an $r < 1$, which is a *constant*, such that the ratio of all pairs of consecutive terms never exceeds r . In the harmonic series, no such r exists because the ratio becomes arbitrarily close to 1.

Splitting summations

One way to obtain bounds on a difficult summation is to express the series as the sum of two or more series by partitioning the range of the index and then to bound each of the resulting series. For example, let's find a lower bound on the arithmetic series $\sum_{k=1}^n k$, which we have already seen has an upper bound of n^2 . You might attempt to bound each term in the summation by the smallest term, but since that term is 1, you would get a lower bound of n for the summation—far off from the upper bound of n^2 .

You can obtain a better lower bound by first splitting the summation. Assume for convenience that n is even, so that

$$\begin{aligned}\sum_{k=1}^n k &= \sum_{k=1}^{n/2} k + \sum_{k=n/2+1}^n k \\ &\geq \sum_{k=1}^{n/2} 0 + \sum_{k=n/2+1}^n \frac{n}{2} \\ &= \left(\frac{n}{2}\right)^2 \\ &= \Omega(n^2),\end{aligned}$$

which is an asymptotically tight bound, since $\sum_{k=1}^n k = O(n^2)$.

For a summation arising from the analysis of an algorithm, you can sometimes split the summation and ignore a constant number of the initial terms. Generally, this technique applies when each term a_k in a summation $\sum_{k=0}^n a_k$ is independent of n . Then for any constant $k_0 > 0$, you can write

$$\begin{aligned}\sum_{k=0}^n a_k &= \sum_{k=0}^{k_0-1} a_k + \sum_{k=k_0}^n a_k \\ &= \Theta(1) + \sum_{k=k_0}^n a_k,\end{aligned}$$

since the initial terms of the summation are all constant and there are a constant number of them. You can then use other methods to bound $\sum_{k=k_0}^n a_k$. This technique applies to infinite summations as well. For example, let's find an asymptotic upper bound on $\sum_{k=0}^{\infty} k^2/2^k$. The ratio of consecutive terms is

$$\frac{(k+1)^2/2^{k+1}}{k^2/2^k} = \frac{(k+1)^2}{2k^2} \leq 8/9$$

if $k \geq 3$. Thus, you can split the summation into

$$\begin{aligned} \sum_{k=0}^{\infty} \frac{k^2}{2^k} &= \sum_{k=0}^2 \frac{k^2}{2^k} + \sum_{k=3}^{\infty} \frac{k^2}{2^k} \\ &= \sum_{k=0}^2 \frac{k^2}{2^k} + \sum_{k=0}^{\infty} \frac{(k+3)^2}{2^{k+3}} \quad (\text{by reindexing}) \\ &\leq \sum_{k=0}^2 \frac{k^2}{2^k} + \frac{9}{8} \sum_{k=0}^{\infty} \left(\frac{8}{9}\right)^k \quad (\text{by inequality (A.15)}) \\ &= (0 + 1/2 + 1) + \frac{9/8}{1 - 8/9} \quad (\text{by equation (A.16)}) \\ &= O(1). \end{aligned}$$

The technique of splitting summations can help determine asymptotic bounds in much more difficult situations. For example, here is one way to obtain a bound of $O(\lg n)$ on the harmonic series (A.9):

$$H_n = \sum_{k=1}^n \frac{1}{k}.$$

The idea is to split the range 1 to n into $\lfloor \lg n \rfloor + 1$ pieces and upper-bound the contribution of each piece by 1. For $i = 0, 1, \dots, \lfloor \lg n \rfloor$, the i th piece consists of the terms starting at $1/2^i$ and going up to but not including $1/2^{i+1}$. The last piece might contain terms not in the original harmonic series, giving

$$\begin{aligned} \sum_{k=1}^n \frac{1}{k} &\leq \sum_{i=0}^{\lfloor \lg n \rfloor} \sum_{j=0}^{2^i-1} \frac{1}{2^i + j} \\ &\leq \sum_{i=0}^{\lfloor \lg n \rfloor} \sum_{j=0}^{2^i-1} \frac{1}{2^i} \\ &= \sum_{i=0}^{\lfloor \lg n \rfloor} \left(2^i \cdot \frac{1}{2^i} \right) \\ &= \sum_{i=0}^{\lfloor \lg n \rfloor} 1 \\ &\leq \lg n + 1. \end{aligned} \tag{A.17}$$

Approximation by integrals

When a summation has the form $\sum_{k=m}^n f(k)$, where $f(k)$ is a monotonically increasing function, you can approximate it by integrals:

$$\int_{m-1}^n f(x) dx \leq \sum_{k=m}^n f(k) \leq \int_m^{n+1} f(x) dx . \quad (\text{A.18})$$

Figure A.1 justifies this approximation. The summation is represented as the area of the rectangles in the figure, and the integral is the blue region under the curve. When $f(k)$ is a monotonically decreasing function, you can use a similar method to provide the bounds

$$\int_m^{n+1} f(x) dx \leq \sum_{k=m}^n f(k) \leq \int_{m-1}^n f(x) dx . \quad (\text{A.19})$$

The integral approximation (A.19) can be used to prove the tight bounds in inequality (A.10) for the n th harmonic number. The lower bound is

$$\begin{aligned} \sum_{k=1}^n \frac{1}{k} &\geq \int_1^{n+1} \frac{dx}{x} \\ &= \ln(n+1) , \end{aligned} \quad (\text{A.20})$$

For the upper bound, the integral approximation gives

$$\begin{aligned} \sum_{k=1}^n \frac{1}{k} &= \sum_{k=2}^n \frac{1}{k} + 1 \\ &\leq \int_1^n \frac{dx}{x} + 1 \\ &= \ln n + 1 . \end{aligned} \quad (\text{A.21})$$

Exercises

A.2-1

Show that $\sum_{k=1}^n 1/k^2$ is bounded above by a constant.

A.2-2

Find an asymptotic upper bound on the summation

$$\sum_{k=0}^{\lfloor \lg n \rfloor} \lceil n/2^k \rceil .$$

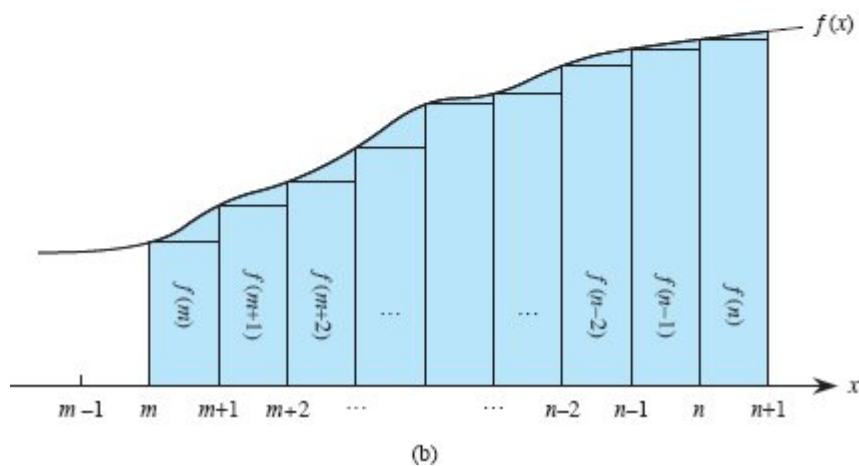
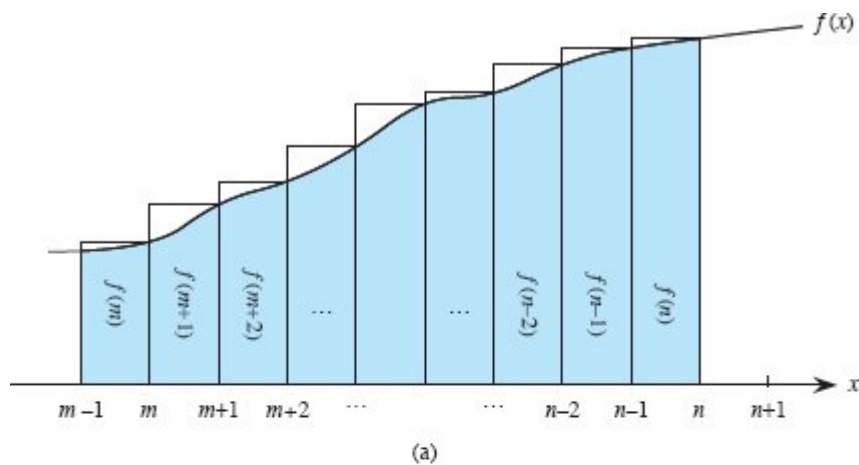


Figure A.1 Approximation of $\sum_{k=m}^n f(k)$ by integrals. The area of each rectangle is shown within the rectangle, and the total rectangle area represents the value of the summation. The integral is represented by the blue area under the curve. Comparing areas in **(a)** gives the lower bound $\int_{m-1}^n f(x) dx \leq \sum_{k=m}^n f(k)$. Shifting the rectangles one unit to the right gives the upper bound $\sum_{k=m}^n f(k) \leq \int_m^{n+1} f(x) dx$ in **(b)**.

A.2-3

Show that the n th harmonic number is $\Omega(\lg n)$ by splitting the summation.

A.2-4

Approximate $\sum_{k=1}^n k^3$ with an integral.

A.2-5

Why can't you use the integral approximation (A.19) directly on $\sum_{k=1}^n 1/k$ to obtain an upper bound on the n th harmonic number?

Problems

A-1 Bounding summations

Give asymptotically tight bounds on the following summations. Assume that $r \geq 0$ and $s \geq 0$ are constants.

a. $\sum_{k=1}^n k^r.$

b. $\sum_{k=1}^n \lg^s k.$

c. $\sum_{k=1}^n k^r \lg^s k.$

Appendix notes

Knuth [259] provides an excellent reference for the material presented here. You can find basic properties of series in any good calculus book, such as Apostol [19] or Thomas et al. [433].